Brent Philip Perdew

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OBJECTIVE

Entry-level engineer seeking to apply my background in aerospace systems, modeling and simulation, and software development to support complex engineering challenges in defense and space operations environments.

EDUCATION

Georgia Institute of Technology | Atlanta, GA — August 2022 - Present

Bachelor of Science in Aerospace Engineering, Overall GPA- 3.50 — Class of 2026

Physics Minor

WORK EXPERIENCE

Voyager Space — Houston, Texas

Engineering Intern – Systems Engineering — May 2024 - August 2024

- Conducted trade studies on commercial space research platforms supporting ISS, Starlab, and Gateway architectures, evaluating structural, power, and interface compatibility for hosted payload integration
- Developed a decision framework for assessing system-level constraints across multiple orbital platforms
- Collaborated with engineers and stakeholders to communicate technical findings through formal design presentations

PROJECT EXPERIENCE

Autonomous On-Orbit Satellite Servicing Mission — Atlanta, Georgia

Space System Design — January – May 2026

- Leading system-level architecture and requirements development for a GEO satellite servicing vehicle designed to service, repair, and refuel LM2100 satellite buses
- Serve as Project Systems Engineer Lead coordinating subsystem interfaces, requirements flow-down, and verification planning
- Contributing to Guidance, Navigation, and Control design for rendezvous and proximity operations, including attitude control and orbital maneuver considerations

Thermal Protection Systems Computing Project — Atlanta, Georgia

Aerothermodynamics — August - December 2025

- Developed a modular MATLAB simulation for a truncated elliptic cone hypersonic vehicle, computing aerodynamic coefficients across Mach 10–35 using modified Newtonian theory and normal/oblique shock relations, and evaluating stagnation and surface heating along a prescribed trajectory
- Integrated transient heat flux to obtain total heat loads and performed margin-based thermal protection system sizing using finite-difference thermal modeling to determine required multi-layer TPS thicknesses under trajectory dispersion and heating uncertainty.

Jet Engine Design — Limerick, Ireland

Jet & Rocket Propulsion — June - July 2025

- Optimized jet engine performance based on mission inputs including flight duration and target Mach number
- Developed an algorithm to identify the highest performing jet engine configuration for a given set of parameters

RELEVANT COURSEWORK

Aerothermodynamics, Aerodynamics, Fluids and Solids Lab, Controls Systems, System Dynamics, Structural Analysis, Mathematical Methods in Engineering, Classical Mechanics, Space System Design, Orbital Mechanics

SKILLS & CERTIFICATIONS

MATLAB/Simulink, Python, GMAT, SolidWorks, GNC, Systems Engineering, Mission Architecture, Eagle Scout

CAMPUS & COMMUNITY INVOLVEMENT

Executive Board, *Tau Kappa Epsilon* — August 2023 – April 2026

- Planned and ran philanthropy, alumni networking, new member education/initiation, and mentoring events